Example 4.21 P and Q are the mid-points of the sides CA and CB respectively of a ΔABC, right angled at C. Prove that $4(AQ^2 + BP^2) = 5AB^2$.

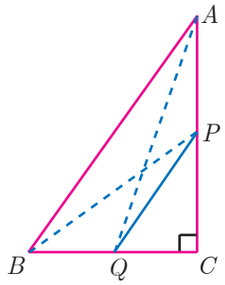


Fig. 4.48

Solution ΔAQC is a right triangle at C, $AQ^2 = AC^2 + QC^2$ ... (1)
ΔBPC is a right triangle at C, $BP^2 = BC^2 + CP^2$ ... (2)
ΔABC is a rignt triangle at C, $AB^2 = AC^2 + BC^2$ ... (3)

From (1) and (2), $AQ^2 + BP^2 = AC^2 + QC^2 + BC^2 + CP^2$

$$4(AQ^2 + BP^2) = 4AC^2 + 4QC^2 + 4BC^2 + 4CP^2$$
$$= 4AC^2 + (2QC)^2 + 4BC^2 + (2CP)^2$$
$$= 4AC^2 + BC^2 + 4BC^2 + AC^2 \quad \text{(Since } P \text{ and } Q \text{ are mid points)}$$
$$= 5(AC^2 + BC^2) \quad \text{(From equation (3))}$$
$$4(AQ^2 + BP^2) = 5AB^2$$

Example 4.22 What length of ladder is needed to reach a height of 7 ft along the wall when the base of the ladder is 4 ft from the wall? Round off your answer to the next tenth place.

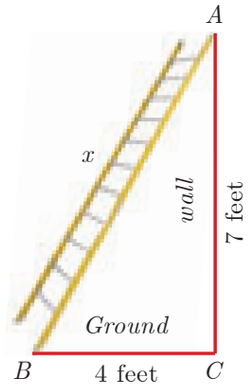


Fig. 4.49

Solution Let x be the length of the ladder. $BC = 4$ ft, $AC = 7$ ft.

By Pythagoras theorem we have, $AB^2 = AC^2 + BC^2$

$x^2 = 7^2 + 4^2 \Rightarrow x^2 = 49 + 16$

$x^2 = 65$. Hence, $x = \sqrt{65}$

The number $\sqrt{65}$ is between 8 and 8.1.

$8^2 = 64 < 65 < 65.61 = 8.1^2$

Therefore, the length of the ladder is approximately 8.1 ft.

Example 4.23 An Aeroplane after take off from an airport and flies due north at a speed of 1000 km/hr. At the same time, another aeroplane take off from the same airport and flies due west at a speed of 1200 km/hr. How far apart will be the two planes after 1½ hours?

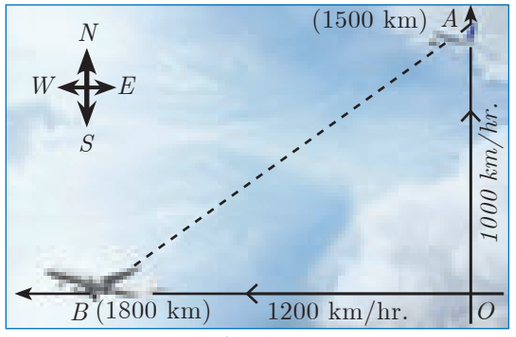


Fig. 4.50

Solution Let the first aeroplane starts from O and goes upto A towards north, (Distance = Speed × time)

where $OA = \left(1000 \times \frac{3}{2}\right)$ km $= 1500$ km

Let the second aeroplane starts from O at the same time and goes upto B towards west,

where $OB = \left(1200 \times \frac{3}{2}\right) = 1800$ km

The required distance to be found is BA.

In right angled tirangle AOB, $AB^2 = OA^2 + OB^2$

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